

M E R N Define Problem Statements

shopEZ — B2C E-Commerce Marketplace (MERN Stack)

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Define Problem Statements

1. Introduction

This document formalises the problem statements identified during the Brainstorming & Ideation phase. Each statement follows the structured "I am / I'm trying to / But / Because / Which makes me feel" framework to ensure user-centric problem articulation. Root-cause analysis is subsequently applied to identify systemic factors driving these problems.

2. Problem Statements

PS-1: The Online Shopper

Field	Description
I am	A busy online shopper seeking electronics and clothing items
I'm trying to	Find and purchase specific items quickly and securely from a trusted platform
But	Existing stores suffer from poor search functionality, confusing checkouts, and loss of cart state
Because	Their systems are inadequately optimised - they lack advanced filtering, cross-device cart persistence, and payment integrations
Which makes me feel	Frustrated, distrustful, and highly likely to abandon my cart prior to purchase

3. Root-Cause Analysis (Ishikawa Diagram)

Poor E-commerce Experience

- Platform
- User Experience
- Technology
- Security
- No SPA / Slow Page Reloads
- No Persistent Cart State
- Limited Customisability
- Confusing Checkout Flow
- Poor Search & Filter Capabilities
- Non-Responsive Mobile Layout
- No CDN for Image Delivery
- No Real-Time State Updates
- Inadequate Admin Dashboard
- No JWT / Stateless Auth
- No Trusted Payment Gateway
- Unencrypted Password Storage

FIGURE 2.1 — ISHIKAWA (FISHBONE) ROOT-CAUSE DIAGRAM

4. Problem-to-Feature Mapping

Root Cause	Proposed Solution Feature
No SPA / Slow page reloads	React.js Single Page Application with Virtual DOM
No persistent cart state	MongoDB-backed cart schema linked to authenticated user session
Poor search & filter	Full-text search with price, category, and rating filters
Confusing checkout	Guided 3-step checkout wizard (Address -> Method -> Payment)
No CDN for images	Cloudinary integration for optimised image delivery
No trusted payment gateway	Stripe / Razorpay payment gateway integration
Unencrypted passwords	Bcrypt password hashing before persistence
No JWT auth	JWT tokens stored in HTTP-only cookies for XSS prevention
No admin dashboard	Dedicated /admin panel with sales analytics and order management

5. Conclusion

The two problem statements — representing the end consumer and the store operator — collectively define the dual-focus of the ShopEZ platform. All identified root causes have corresponding solution features mapped in the Proposed Solution and Requirement Analysis documents.
